

- Q.29 Define Slip and slip frequency of rotor of a three phase Induction Motor.
- Q.30 Explain the working principal of Synchronous motor.
- Q.31 Explain the working principal of Linear Induction motor. Also write its applications
- Q.32 Explain the concept of Armature reaction in Synchronous Machine in Detail.
- Q.33 Write down the conditions for parallel operation of alternators.
- Q.34 What are the various applications of Synchronous motor?
- Q.35 Compare Squirrel cage and Phase wound Induction motor.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain different starting methods of Induction Motor.
- Q.37 Explain the construction and working of universal motor with the help of diagram.
- Q.38 Explain the construction and working of Induction motor with the help of diagram

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5th Semester/ Elect, Power Station Engg., Elect. & Eltx. Engg. Sub. :Electrical Machines-II

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A motor in which the rotor turns in discrete movements is called
a) Servo motor b) Stepper motor
c) Universal motor d) All of these
- Q.2 The term cogging is associated with _____ motor.
a) Induction motor b) Synchronous motor
c) DC Motor d) None of these
- Q.3 The Induction motor shaft is made of
a) Mild steel b) Stainless steel
c) Cast Iron d) All of these
- Q.4 The rating of alternators is usually expressed in
a) Ampere b) Volt
c) KVA d) KW
- Q.5 Maximum speed of a synchronous machine for 50Hz supply frequency is ____ R.P.M.
a) 1500 b) 3000
c) 2000 d) 300

- Q.6 Brushes of phase wound Induction motor are made of
 a) Carbon b) Cast Iron
 c) Steel d) Copper
- Q.7 The machine that supplies DC to the rotor of an alternator is called
 a) Rectifier b) Exciter
 c) Convertor d) Inverter
- Q.8 If the Induction motor is running at synchronous speed, then the slip is
 a) Zero b) One
 c) Infinity d) None of the above
- Q.9 Frequency of AC supply in India is
 a) 60 Hz b) 0Hz
 c) 50Hz d) 25 Hz
- Q.10 Three phase Induction motor are
 a) Self starting
 b) Not self starting
 c) Need D.C. excitation
 d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Synchronous machine runs at _____ speed.
- Q.12 The cage of rotor is made up of copper. (True/False)
- Q.13 Induction motor cannot rotate at Synchronous speed. (True/False)
- Q.14 Hunting in a synchronous motor can be minimized by using_____.

(2)

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- Q.15 The rating of alternators is usually expressed in KW. (True/False)
- Q.16 An overexcited motor working at no load is called_____.
- Q.17 Rotor of a synchronous motor is connected to _____ Supply.
- Q.18 Synchronous motors are inherently self starting. (True/False)
- Q.19 The direction of rotation of a 3 phase Induction Motor can be reversed by_____
- Q.20 Single phase induction motor is self starting. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Drive an expression for induced emf of an Alternator.
- Q.22 Discuss the causes of hunting.
- Q.23 Draw and explain Torque-slip curve of a Induction motor.
- Q.24 Write a short note on stepper motor.
- Q.25 What are the advantages of Squirrel cage three phase Induction motor?
- Q.26 Draw and explain V-Curve for Synchronous motor.
- Q.27 How does a synchronous motor starts?
- Q.28 Compare an Induction motor with Synchronous motor on the basis of starting, cost and efficiency.

(3)

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